METRIC

DOD-P-63981(AR)
AMENDMENT 8
26 August 1992
SUPERSEDING
AMENDMENT 7
18 January 1991

MILITARY SPECIFICATION

PROJECTILE 30MM, HIGH EXPLOSIVE DUAL PURPOSE, M789, METAL PARTS

This Amendment forms a part of Military Specification DOD-P-63981(AR), dated 29 October 1982, and is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1, STANDARDS: Add "MIL-STD-1235 Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes".

2.1.2, Drawings: Delete "11825929" and substitute "11825955"

PAGE 3

3.4: Add the following after "...shall fill the band seat.":

"When sectioned along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut areas. Failure of either of the exposed sections to comply with 65% fill in the undercut areas shall be considered a defect"

3.5: Delete in its entirety and substitute the following:

"3.5 Soundness. The projectile bodies shall be examined for metal discontinuities by eddy current or ultrasonic inspection. A discontinuity in excess of .25mm deep by 6.35mm in length shall be classified as defect."

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3.6.1:

Add to end of second sentence: "except coating weight shall be 25mg/sq ft minimum"

Add to end of third sentence: "except 3 panels are required at the beginning of each shift only."

3.6.3: Delete in its entirety and substitute the following:

"3.6.3 Corrosion resistance. The Bourrelet and Ogive portion (exclude area aft of rotating band from the test) of the exterior painted surface of the projectile metal parts shall show no evidence of corrosion (red rust) after exposure to a 5% solution of atomized salt spray for 24 hours."

PAGE 4

* 3.7: Should read as follows:

"3.7 Metal Parts Security. The projectile metal parts when inert charged, fuzed and assembled into test cartridges meeting the ballistic requirements of Dwg. 11825954, shall not separate or break-up when tested for metal parts security."

PAGE 5

4.3.1: Delete Quantity "125" and substitute "64", Quantity "400" and substitute "560".

PAGE 6

Table I:

Delete all notes at the bottom of the page and all reference to them.

Band tightness: Delete "125" and substitute "32".

Projectile, Metal Parts, Examination for defects: Delete "32" and substitute "32*".

Projectile, Metal Parts, Soundness: Delete "125" and substitute "125*".

Notes: Add "* This inspection can be performed on the same sample".

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PAGE 7

4.4.2.a: Delete "unless otherwise specified" at the end of the paragraph and substitute "or MIL-STD-1235".

PAGE 8

4.4.2.1 Delete "minor 301, width, maximum .40%, 3.3.1 Gage".

PAGE 9

4.4.2.2, Major 105: Delete "base".

PAGE 11

4.4.2.3:

Major 102: Delete "min" and substitute "max".

Major 104: Delete "runout" and substitute "true position".

Major 106, Change From:

Major 106.	Band tightness	20	Acc-0	3.4	4.5.1

TO:

Major 106.	Band tightness	20	Acc-0 Rej-1	3.4	4.5.1/Gage

Notes: Delete "CD 108 and 110" and substitute "Major 106, 113 and Minor 203."

Delete: "(See Note)" from major 106.

Add: "(See Note)" under title.

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4.4.2.4:

Minor 201, Paragraph Reference/Inspection Method column add "6.8."

Add "11825956" to the next higher assembly.

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4.4.3.1:

Delete "Sand fill" and substitute "Band fill".

Delete the last sentence and substitute "examination per 4.5.1".

4.4.3.2, Second sentence: Delete "flow" and substitute "flaw".

Delete paragraph 4.5.1 in its entirety and replace with the following:

"4.5.1 Band tightness. After sectioning along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut areas using a .400mm No-go pin. Failure of either of the exposed sections to comply with the 65% fill requirement shall be considered a defect."

4.5.2.1: Delete in its entirety.

4.5.3, First sentence: Delete "at the start and completion of".

4.5.4: Delete second sentence and substitute with "The prescribed panels shall be processed with the projectiles to be represented, per paragraph 3.6.1".

4.5.5: Delete "XM230" and substitute "M230".

4.5.7:

Second and third sentence: Delete "target" and substitute "witness panel".

Second sentence: Delete "8'x8'1"" and substitute "8'x8'xl"".

* Fourth sentence to read as follows:

"The test sample shall be assembled to comply with the ballistic requirements of Dwg. 11825954, Cartridge, 30MM, for projectile metal parts test."

Fifth sentence: Delete "bursts of 25 shots" and substitute "20 round minimum bursts".

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6.8: Add "d" as follows:

"d. Missing paint on the rotating band is not a defect."

6.9, Add new paragraph as follows:

"6.9 Intermediate point inspection. The classification of defects identifies the defect characteristics (among other things) for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality in the end item."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

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